

NAYAN GODHWANI

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Los Angeles, CA

EDUCATION

University of Southern California, Los Angeles, CA

Aug 2024-May 2026

Master of Science - Computer Science

GPA: 3.58/4.00

Coursework: Analysis of Algorithms, Machine Learning for Data Science, Operating Systems, Applied NLP

Shri Ramdeobaba College of Engineering and Management, Maharashtra, India.

Aug 2018-Jun 2022

Bachelor of Engineering - Computer Science

GPA: 3.7/4.0

WORK EXPERIENCE

Software Engineer Intern

Feb 2025-May 2026

USC Alzheimer's Therapeutic Research Institute, San Diego, CA

Electronic Data Capture System

- Enhanced EDC system for Alzheimer's clinical trials built on **Django ORM** and **ReactJS** to monitor **4K+** trial participants.
- Engineered a scalable REST API with **Redis caching** using **proxy design pattern** to serve **2K+** users globally with **7K+** PDF downloads, cutting median response time by **42%**.
- Redacted PHI across **100K+** clinical-trial records to ensure HIPAA compliance and protect sensitive patient data using automated field-level masking logic.

AI Intern

Jun 2025-Aug 2025

Lumina View, San Francisco, CA

- Fine-tuned a local LLaMA model on RecipeNLG (**2.3M+** recipes) using HuggingFace and Pytorch, deployed with CrewAI agents to generate personalized recipes based on user preferences.
- Implemented RAG-based retrieval with ChromaDB vector embeddings and **NVIDIA NeMo** guardrails to maintain safe and accurate responses.
- Executed **50** CrewAI evaluation tests to report an F1 score of **0.85**.

Software Engineer

Aug 2022-Jun 2024

HashedIn By Deloitte, Maharashtra, India

NextGen Point of Sale

- Collaborated as a backend developer, for a retail industry, to automate checkout process, enabling **10M+** customers to use self-checkout kiosks for purchase.
- Implemented an event-driven, **MQTT** based, architecture for asynchronous communication across 3 devices with **Spring Boot**, processing **1M+** transactions/day with **99.9%** uptime.
- Designed a **ThreadPoolExecutor** task scheduler to automate **10-second** polling of customer deposits.

Data Migration

- Designed a scalable **ETL** pipeline to process over **5 GB** of data daily, migrating data from Teradata to Snowflake.
- Leveraged a **SQL** stored procedure to create **105** target tables in Snowflake and leveraged AWS Lambda function to deploy **11** permission-based views.
- Received **Spot Award** for implementing an **AWS Lambda** function in **Python** to merge and sort **5 GB** data.
- Validated migration of **1M+** rows across Teradata and Snowflake, ensuring 95% consistency.

Mental Health Awareness

- Spearheaded as a full stack developer to a mental health mobile application that facilitate **500** users in identifying illness and connecting with therapist.
- Led the development of backend REST APIs, using **Android(Java)**, for user management, achieving **20%** latenc reduction through caching.
- Collaborated with UI designer to enhance **8** mobile screens, contributing to **25%** increase in active users.

SKILLS

Programming Languages: Java 8, Python, C/C++, Javascript

Web/Backend: Spring, Spring Boot, Django, HTML, CSS, ReactJS, NodeJS

Data/Cloud: SQL, MySQL, PostgreSQL, Redis, Linux, HuggingFace, PyTorch, Tensorflow,scikit-learn

PROJECTS

Weenix Operating System

- Developed Unix OS kernel components in C++, focusing on process management, thread scheduling and synchronization.
- Implemented a modular virtual file system (VFS), optimizing extensible file operations and device I/O management.
- Enhanced the virtual memory subsystem by implementing page fault handling, copy-on-write, system call support mmap(), read(), write()).

Network Packet Transmission Simulator with Multithreading

- Devised a multithreaded network packet transmission simulator in C using pthreads, condition variables, and synchronization primitives.
- Coordinated 4 concurrent threads (packet arrival, token generation, and two parallel servers) to simulate real-world scheduling, ensuring safe thread synchronization, deadlock prevention, and efficient throughput under varying load conditions.